$\langle qu(b(i(t|t))ick)\rangle$ — the four-era canon

Calibration Policy

The Declared Slots of the Corpus

Companion document to the four-era canon, Volume I

Every calibration slot, what it calibrates, which results depend on it, and its status under P5

Edition of 2026-10-09

Calibration is a policy decision, not a mathematical derivation; this document therefore stands apart from the monograph. All slot data are transcribed from the corpus's own registries (Part 1 §E; Part 2 §Cal-2; Part 3 §Cal-F; Part 4 §Cal-BM).

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Purpose and standing

This document lists every calibration slot of the corpus. A slot is a symbol the corpus declares as free internal configuration data (P4-class) or logs as a spectral datum, without issuing a numerical value. The governing constraints are the policy clauses P1–P5 of Part 1 §E, reproduced verbatim below; in particular:

- P1 — $\exists!$ free dimensionful scale: $\hbar$; every other quantity is dimensionless-derived or a power of $\hbar$.
- P4 — free internal configuration parameters are $\neg$constants of the theory; predictions $:=$ dimensionless ratios among derived quantities at fixed core data.
- P5 — $\neg\exists$ quantity identified with a measured external constant; comparison is a test of A9, $\neg$ an input to A1–A8.

Every slot below is declared but $\neg$issued: the corpus fixes its locus, its dimension and its consumers, and no theorem, audit or annex attaches a numerical value to it. No slot enters any machine-verified identity as a number; the verification suite (`verification/run_all.py`) is unaffected by any future issuance, which would itself require a ledger addendum first.

The policy clauses P1–P5 (Part 1 §E, verbatim)

P1. $\forall$ quantities of the corpus: derived from A1–A9 $\wedge$ C1–C13 $\wedge$ D0.1–D4.3, except the hop length $\hbar$; $[\hbar] = \mathfrak{q}^{-1/2}$; $\hbar$ fixes the absolute scale.

P2. Derived, zero freedom:

$$(\dim \mathcal{E}_{++}, \dim \mathcal{E}_{+-}, \dim \mathcal{E}_{-+}, \dim \mathcal{E}_{--}) = (7,3,3,3); \quad C_{\text{cap}} = 16; \quad D_{\text{tick}}(n) = 2^{n+1}; \quad n_{\max} = 3;$$

$$N_{\text{colour}} = 3; \quad N_{\text{gluon}} = 8; \quad N_{\gamma\text{-pol}} = 2; \quad q_{\text{EM}} \in \{-1, 0, +1\}; \quad \alpha_e \in \{0, \pi\} \text{ per edge};$$

$$m_W = m_Z \cos\theta_W; \quad e = g \sin\theta_W; \quad S = n_{\mathcal{R}} \ln 2; \quad w_{D^+} = -1.$$

Audit note. Every entry of this list is re-verified from first principles in exact arithmetic: `checks/verify_merge_tierB.py` §F (totals and cross-references; the identity-level checks live in `verify_B.py`–`verify_G.py`).

P3. Named combinations of $\mathfrak{h}$ (external comparison only; ¬ inputs):

$$G_N = \frac{\mathfrak{h}^2}{4 \ln 2}; \qquad \kappa^2 = 8\pi G_N = \frac{2\pi\, \mathfrak{h}^2}{\ln 2} \qquad [\text{CT2.9}].$$

P4. Free internal configuration parameters (¬ constants of the theory). Independent core: g, g'; h_0; c_W; Λ; $\varkappa$, β, α_Ω. Functionally dependent members (properties of a solution, ¬ further freedom): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; m_W, m_Z [EW.20, EW.23]; m_H [EW.27]. Predictions := dimensionless ratios among derived quantities at fixed core data.

P5. ¬∃ quantity identified with a measured external constant. Comparison procedure: fix $\mathfrak{h}$ once; read P2 ∧ P3; result := test of A9, ¬ input to A1–A8. Era-extension note. The P4 core lists the bosonic-era free internal parameters (Part 2 loci). The fermion-era parameters (the Janus rate ratio $r = x_+/x_-$; the defect residues $\hat{\mu}^2$, $\hat{\mu}'^2$; the Ansatz-B formal temperature α; the channel scales μ, μ_ν; the formal fugacities) are registered in Part 3 §Cal-F under this policy; the baryomesonic era introduces none (Part 4 §Cal-BM).

The consolidated slot register

One row per slot; the sheets that follow give the full record. P4-class members are free internal configuration parameters (¬constants of the theory); all values are ¬issued [P5].

Slot	Calibrates	Registration	Status
$\mathfrak{h}$	the hop length — the single free dimensionful scale; $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$; fixes the absolute scale (P1; D0.6)	Part 1 §E (P1, P3); D0.6; D2.9	free scale, value ¬issued [P1, P5]
g	the weak coupling (§D: weak, hypercharge, EM couplings)	Part 1 §E (P4 core); EW.33	declared, value ¬issued [P4, P5]
g'	the hypercharge coupling (§D: weak, hypercharge, EM couplings)	Part 1 §E (P4 core); EW.33	declared, value ¬issued [P4, P5]
h_0	the Higgs background (§D)	Part 1 §E (P4 core); EW.33	declared, value ¬issued [P4, P5]
c_W	the electroweak mixing datum entering the scalar trace mode	Part 1 §E (P4 core); EW.24, EW.33	declared, value ¬issued [P4, P5]
Λ	the stretch scalar — identity on the D^+ force space ($F_{D^+} = \Lambda\,\mathbb{I}$) [A8; §D]	Part 1 §E (P4 core); A8; §D	declared, value ¬issued [P4, P5]
$\varkappa$	the sector-to-metric embedding scale [C4; D2.1]; $\varkappa \neq \kappa$ (C4)	Part 1 §E (P4 core); C4; D2.1	declared, value ¬issued [P4, P5]
β	a metric embedding constant [§D; D2.1]	Part 1 §E (P4 core); D2.1	declared, value ¬issued [P4, P5]

Slot	Calibrates	Registration	Status
α_Ω	a metric embedding constant [§D; D2.1]	Part 1 §E (P4 core); D2.1	declared, value ¬issued [P4, P5]
$\alpha_{\mathrm{EM}}^{(2)}$	the EM-block scale read [EM.17] — a spectral datum of F at the solution level, ¬a fitted constant and ¬a P4 core member (Cal-2)	Part 2 §Cal-2; EM.17; EM.25(8)	logged spectral datum; value ¬issued; any future use that needs a value requires a HistLog §I addendum entry first (Cal-2)
ρ_{B}	the bridge's dimensionless calibration ratio $\rho_{\mathrm{B}} = \hat{\mu}'^2\mathfrak{h}^2/(\ln 2)^2 \in \mathfrak{q}^0$ — a packaging of Cal-F's $\hat{\mu}'^2$ and P1's $\mathfrak{h}$, hence ¬a new free parameter and ¬a new dimensionful scale (C-BM.1)	Part 4 §Cal-BM (C-BM.1; A-BM.9, Tier BM.20); BM.143(4) consolidation	slot declared, value ¬issued [A-BM.9; P5]
e_0	the numerical value of the electric coupling (C-BM.2)	Part 4 §Cal-BM (C-BM.2); EM.25(7)	value ¬issued; ¬fixed by measurement [P5]
$\mu_\nu,\ \mu,\ \rho_\ell,\ \rho_\nu$	the channel scales, inherited from Cal-F, ¬declared in Part 4 (C-BM.3)	Part 4 §Cal-BM (C-BM.3); Part 3 §Cal-F	inherited; values ¬issued [P5]
$r = x_+/x_-$ (Janus rate ratio; equivalently $x_\pm$)	the flicker ensemble; the equilibrium form $p = x_+/(x_+ + x_-)$ is derived [FT.108(4)], the ratio is the datum	Part 3 §Cal-F	declared, value ¬issued [P4, P5]
$\hat{\mu}^2$	the leading-order pole read [FT.33]	Part 3 §Cal-F	declared, value ¬issued [P4, P5]
$\hat{\mu}'^2$	the exact pole read [FT.54(4); Census FT.76]	Part 3 §Cal-F	declared, value ¬issued [P4, P5]
α (Ansatz-B formal temperature)	class masses $m_B = (K - \alpha \ln \Omega) \ln 2/\mathfrak{h}$; no value declared; Ansatz-B results stand as exact (K, a, b) classes	Part 3 §Cal-F	declared, value ¬issued [P4, P5]
μ (family-channel overall scale)	channel spectra; every ratio is μ-free	Part 3 §Cal-F	declared, value ¬issued [P4, P5]
μ_ν (rung-one channel scale)	the rung-one spectrum; ratios exact, scale undeclared	Part 3 §Cal-F	declared, value ¬issued [P4, P5]
$t, u, x_\pm, y_d, z_d, \tau_{\mathcal{R}}, \ldots$ (formal fugacities)	formal indeterminates over $\mathbb{Q}$; specializations marked at use [FT.108(4); Cor L0a]	Part 3 §Cal-F	declared, value ¬issued [P4, P5]

Slot sheets

Slot $\mathfrak{h}$

- Name. $\mathfrak{h}$ — registered at Part 1 §E (P1, P3); D0.6; D2.9.
- Calibrates. the hop length — the single free dimensionful scale; $[\mathfrak{h}] = \mathfrak{q}^{-1/2}$; fixes the absolute scale (P1; D0.6).
- Dependents. every dimensionful quantity of the corpus is a power of $\mathfrak{h}$ (P1); named combi-

nations $G_N = \mathfrak{h}^2/(4\ln 2)$, $\kappa^2 = 8\pi G_N = 2\pi\mathfrak{h}^2/\ln 2$ [P3; CT2.9].

- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. free scale, value $\neg$issued [P1, P5].

Slot g

- Name. g — registered at Part 1 §E (P4 core); EW.33.
- Calibrates. the weak coupling (§D: weak, hypercharge, EM couplings).
- Dependents. $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $m_W,\ m_Z$ [EW.20, EW.23] (functionally dependent members (properties of a solution, $\neg$ further freedom [P4]): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; $m_W,\ m_Z$ [EW.20, EW.23]; m_H [EW.27]).
- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. declared, value $\neg$issued [P4, P5].

Slot g'

- Name. g' — registered at Part 1 §E (P4 core); EW.33.
- Calibrates. the hypercharge coupling (§D: weak, hypercharge, EM couplings).
- Dependents. $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35] (functionally dependent members (properties of a solution, $\neg$ further freedom [P4]): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; $m_W,\ m_Z$ [EW.20, EW.23]; m_H [EW.27]).
- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. declared, value $\neg$issued [P4, P5].

Slot h_0

- Name. h_0 — registered at Part 1 §E (P4 core); EW.33.
- Calibrates. the Higgs background (§D).
- Dependents. $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; m_H [EW.27] (functionally dependent members (properties of a solution, $\neg$ further freedom [P4]): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; $m_W,\ m_Z$ [EW.20, EW.23]; m_H [EW.27]).
- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. declared, value $\neg$issued [P4, P5].

Slot c_W

- Name. c_W — registered at Part 1 §E (P4 core); EW.24, EW.33.
- Calibrates. the electroweak mixing datum entering the scalar trace mode.

- Dependents. $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; $\theta_W(g, g')$ [EW.24] (functionally dependent members (properties of a solution, ¬ further freedom [P4]): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; m_W, m_Z [EW.20, EW.23]; m_H [EW.27]).
- P5 constraint. ¬∃ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 ∧ P3; result := test of A9, ¬ input to A1–A8 [P5].
- Status. declared, value ¬issued [P4, P5].

Slot Λ

- Name. Λ — registered at Part 1 §E (P4 core); A8; §D.
- Calibrates. the stretch scalar — identity on the D^+ force space ($F_{D^+} = \Lambda\,\mathbb{I}$) [A8; §D].
- Dependents. no dependent member is listed in P4; the D^+ sector datum [A8].
- P5 constraint. ¬∃ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 ∧ P3; result := test of A9, ¬ input to A1–A8 [P5].
- Status. declared, value ¬issued [P4, P5].

Slot $\varkappa$

- Name. $\varkappa$ — registered at Part 1 §E (P4 core); C4; D2.1.
- Calibrates. the sector-to-metric embedding scale [C4; D2.1]; $\varkappa \neq \kappa$ (C4).
- Dependents. $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33] (functionally dependent members (properties of a solution, ¬ further freedom [P4]): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; m_W, m_Z [EW.20, EW.23]; m_H [EW.27]).
- P5 constraint. ¬∃ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 ∧ P3; result := test of A9, ¬ input to A1–A8 [P5].
- Status. declared, value ¬issued [P4, P5].

Slot β

- Name. β — registered at Part 1 §E (P4 core); D2.1.
- Calibrates. a metric embedding constant [§D; D2.1].
- Dependents. $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33] (functionally dependent members (properties of a solution, ¬ further freedom [P4]): $\theta_W(g, g')$ [EW.24]; $e(g, g')$ [EW.35]; $\mu_\star^2(c_W, \varkappa, \beta, h_0)$ [EW.33]; m_W, m_Z [EW.20, EW.23]; m_H [EW.27]).
- P5 constraint. ¬∃ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 ∧ P3; result := test of A9, ¬ input to A1–A8 [P5].
- Status. declared, value ¬issued [P4, P5].

Slot α_Ω

- Name. α_Ω — registered at Part 1 §E (P4 core); D2.1.
- Calibrates. a metric embedding constant [§D; D2.1].
- Dependents. no dependent member is listed in P4.

- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. declared, value $\neg$issued [P4, P5].

Slot $\alpha_{\mathrm{EM}}^{(2)}$

- Name. $\alpha_{\mathrm{EM}}^{(2)}$ — registered at Part 2 §Cal-2; EM.17; EM.25(8).
- Calibrates. the EM-block scale read [EM.17] — a spectral datum of F at the solution level, $\neg$a fitted constant and $\neg$a P4 core member (Cal-2).
- Dependents. no result consumes its value — every edge-level phase read uses the registration data $\alpha_e \in \{0, \pi\}$ [Def-EM.20, EM.20] (Cal-2).
- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. logged spectral datum; value $\neg$issued; any future use that needs a value requires a HistLog §I addendum entry first (Cal-2).

Slot ρ_{B}

- Name. ρ_{B} — registered at Part 4 §Cal-BM (C-BM.1; A-BM.9, Tier BM.20); BM.143(4) consolidation.
- Calibrates. the bridge's dimensionless calibration ratio $\rho_{\mathrm{B}} = \hat{\mu}'^2\mathfrak{h}^2/(\ln 2)^2 \in \mathfrak{q}^0$ — a packaging of Cal-F's $\hat{\mu}'^2$ and P1's $\mathfrak{h}$, hence $\neg$a new free parameter and $\neg$a new dimensionful scale (C-BM.1).
- Dependents. the organic-bridge energy audit [BM.69(2); BM.98; BM.102(5)]; no numerical value attaches to it here or anywhere [P5; BM.69(4)'s four grounds].
- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. slot declared, value $\neg$issued [A-BM.9; P5].

Slot e_0

- Name. e_0 — registered at Part 4 §Cal-BM (C-BM.2); EM.25(7).
- Calibrates. the numerical value of the electric coupling (C-BM.2).
- Dependents. the running-coupling completion [EM.25(7); BM.88(8); BM.133(6)].
- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. value $\neg$issued; $\neg$fixed by measurement [P5].

Slot $\mu_\nu,\ \mu,\ \rho_\ell,\ \rho_\nu$

- Name. $\mu_\nu,\ \mu,\ \rho_\ell,\ \rho_\nu$ — registered at Part 4 §Cal-BM (C-BM.3); Part 3 §Cal-F.
- Calibrates. the channel scales, inherited from Cal-F, $\neg$declared in Part 4 (C-BM.3).
- Dependents. the rung-one spectrum [FT.145(3)]; the beta-decay channel reads [BM.83(5),(7)].

- P5 constraint. $\neg\exists$ quantity identified with a measured external constant; comparison procedure: fix $\mathfrak{h}$ once, read P2 $\wedge$ P3; result := test of A9, $\neg$ input to A1–A8 [P5].
- Status. inherited; values ¬issued [P5].

The fermion-era slots (Part 3 §Cal-F, parsed from the registry table)

- $r = x_+/x_-$ (Janus rate ratio; equivalently $x_\pm$) — locus: Def-H.26; FT.68(5); FT.70–FT.71; dimension: $\mathfrak{q}^0$; consumers: the flicker ensemble; the equilibrium form $p = x_+/(x_+ + x_-)$ is derived [FT.108(4)], the ratio is the datum; status: declared, value ¬issued [P4, P5; any numerical use beyond the listed loci requires a HistLog §I addendum entry first].
- $\hat{\mu}^2$ — locus: Def-H.37(1); dimension: $\mathfrak{q}^2$ [EW.33 packaging; D2.9]; consumers: the leading-order pole read [FT.33]; status: declared, value ¬issued [P4, P5; any numerical use beyond the listed loci requires a HistLog §I addendum entry first].
- $\hat{\mu}'^2$ — locus: Def-H.37(2); dimension: $\mathfrak{q}$ [FT.54(4)]; consumers: the exact pole read [FT.54(4); Census FT.76]; status: declared, value ¬issued [P4, P5; any numerical use beyond the listed loci requires a HistLog §I addendum entry first].
- α (Ansatz-B formal temperature) — locus: Def-H.28; dimension: $\mathfrak{q}^0$; consumers: class masses $m_B = (K - \alpha \ln \Omega) \ln 2/\mathfrak{h}$; no value declared; Ansatz-B results stand as exact (K, a, b) classes; status: declared, value ¬issued [P4, P5; any numerical use beyond the listed loci requires a HistLog §I addendum entry first].
- μ (family-channel overall scale) — locus: FT.113; FT.119; dimension: —; consumers: channel spectra; every ratio is μ-free; status: declared, value ¬issued [P4, P5; any numerical use beyond the listed loci requires a HistLog §I addendum entry first].
- μ_ν (rung-one channel scale) — locus: FT.143; dimension: —; consumers: the rung-one spectrum; ratios exact, scale undeclared; status: declared, value ¬issued [P4, P5; any numerical use beyond the listed loci requires a HistLog §I addendum entry first].
- $t, u, x_\pm, y_d, z_d, \tau_\mathcal{R}, \tau_\mathcal{V}$ (formal fugacities) — locus: Def-H.32(3); dimension: $\mathfrak{q}^0$; consumers: formal indeterminates over $\mathbb{Q}$; specializations marked at use [FT.108(4); Cor L0a]; status: declared, value ¬issued [P4, P5; any numerical use beyond the listed loci requires a HistLog §I addendum entry first].

The era registries (verbatim)

Part 2 §Cal-2 - Calibration note (bosonic era)

The P4 core (g, g'; h_0; c_W; Λ; $\varkappa$, β, α_Ω) and P3 (G_N, κ^2 — named combinations of $\mathfrak{h}$) are Part 1 §E. Logged spectral datum of this part: $\alpha_{\mathrm{EM}}^{(2)}$ — the EM-block scale read [EM.17]: operatorially defined; no theorem of the corpus fixes its numerical value under the dynamic-phase regime [EM.25(8)]; it is a spectral datum of F at the solution level, ¬a fitted constant and ¬a P4 core member; no result consumes its value — every edge-level phase read uses the registration data $\alpha_e \in \{0, \pi\}$ [Def-EM.20, EM.20]. Any future use that needs a value requires a logged declaration first.

Part 3 §Cal-F - Free internal parameters of the fermion era (P4-class)

Policy: Part 1 §E (P4: free internal configuration parameters, ¬constants of the theory; P5: no external identification; P1: $\mathfrak{h}$ the single free dimensionful scale). Every entry: dimensionless internal

configuration data unless a dimension is stated; no value is fitted to external data; any numerical use beyond the loci listed requires a logged declaration first.

parameter	locus	dimension	consumers
$r = x_+/x_-$ (Janus rate ratio; equivalently $x_\pm$)	Def-H.26; FT.68(5); FT.70–FT.71	$\mathfrak{q}^0$	the flicker ensemble; the equilibrium form $p = x_+/(x_+ + x_-)$ is derived [FT.108(4)], the ratio is the datum
$\hat{\mu}^2$	Def-H.37(1)	$\mathfrak{q}^2$ [EW.33 packaging; D2.9]	the leading-order pole read [FT.33]
$\hat{\mu}'^2$	Def-H.37(2)	$\mathfrak{q}$ [FT.54(4)]	the exact pole read [FT.54(4); Census FT.76]
α (Ansatz-B formal temperature)	Def-H.28	$\mathfrak{q}^0$	class masses $m_B = (K - \alpha \ln \Omega) \ln 2/\mathfrak{h}$; no value declared; Ansatz-B results stand as exact (K, a, b) classes
μ (family-channel overall scale)	FT.113; FT.119	—	channel spectra; every ratio is μ-free
μ_ν (rung-one channel scale)	FT.143	—	the rung-one spectrum; ratios exact, scale undeclared
$t, u, x_\pm, y_d, z_d, \tau_\mathcal{R}, \tau_\mathcal{V}$ (formal fugacities)	Def-H.32(3)	$\mathfrak{q}^0$	formal indeterminates over $\mathbb{Q}$; specializations marked at use [FT.108(4); Cor L0a]

Part 4 §Cal-BM - Free internal parameters of the baryomesonic era (P4-class)

None introduced by this part. Inherited and undeclared: the family-channel overall scale μ [Cal-F; FT.113, FT.119 — every ratio is μ-free]. Declared slot, value ¬issued (A-BM.9, Tier BM.20): the bridge's dimensionless calibration ratio $\rho_\mathrm{B} = \hat{\mu}'^2\mathfrak{h}^2/(\ln 2)^2 \in \mathfrak{q}^0$ [BM.69(2); BM.98; BM.102(5)] — a packaging of Cal-F's $\hat{\mu}'^2$ and P1's $\mathfrak{h}$, hence ¬a new free parameter and ¬a new dimensionful scale; no numerical value attaches to it here or anywhere [P5; BM.69(4)'s four grounds stand: the slot is declared, ¬derived]. No second dimensionful scale exists [P1; the derived no-go: Part 1 §E]. Any numerical use of an inherited scale beyond its Part-3 loci requires a logged declaration first.

Consolidated calibration register (Tier BM.23, BM.143(4) — a consolidation, ¬an addition; the parameter count is unchanged).

(C-BM.1) ρ_B

$= \hat{\mu}'^2\mathfrak{h}^2/(\ln 2)^2 \in \mathfrak{q}^0$: the bridge ratio; slot declared [A-BM.9],

value ¬issued [BM.143(4)];

(C-BM.2)

e_0 : the numerical value of the electric coupling [EM.25(7); BM.88(8); BM.133(6)];

value ¬issued;

¬fixed by measurement [P5];

(C-BM.3) μ_ν,

$$\mu,$$

$$\rho_\ell,$$

$$\rho_\nu : \quad \text{the channel scales, inherited from Cal-F,}$$

$$\neg\text{declared in this part } [\text{FT.145}(3);\ \text{BM.83}(5), (7)].$$

Each row is a calibration datum of the P4/A-BM.9 species, ¬a structural gap; no row carries a value; ℏ remains the single dimensionful scale [P1].

P5 and the external-data quarantine

> External-data quarantine. The [EXT] audit rows (FT.114, FT.120; and the audit clauses of FT.77, FT.79, FT.87, FT.104 in Part 3) are quarantined comparison data: tested, never trusted; they enter no derivation [P1, P5; FT.73(5) pattern].

External comparison data enter only quarantined audit rows (Census FT.114, FT.120; P1, P5): tested, never trusted, entering no derivation. No slot of this register is fixed by any such row.